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Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

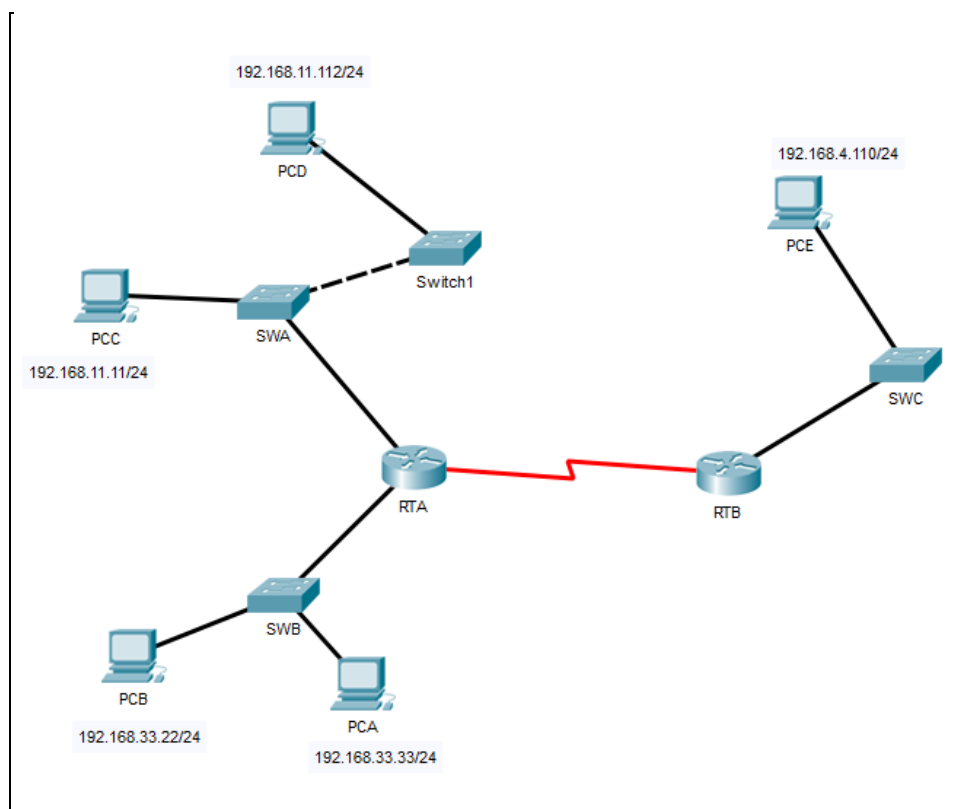


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0002.4a00.0e91    DYNAMIC   Fa0/1
1     000c.8546.7d85    DYNAMIC   Fa1/1
```

SWB

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a    DYNAMIC Fa0/1
```

SWC

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614    DYNAMIC Fa0/1
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type `arp -a` and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA

```
C:\>arp -a
No ARP Entries Found
```

PCB

```
C:\>arp -a
No ARP Entries Found
```

PCC

```
C:\>arp -a
No ARP Entries Found
```

PCD

```
C:\>arp -a
No ARP Entries Found
```

PCE

```
C:\>arp -a
No ARP Entries Found
```

- e. What are your thoughts on the results?

No traffic is being generated between PCs. There is no entry for all PCs because none of them received an ARP request. There will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746        dynamic
```

- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06        dynamic
```

- In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.

PCC

```
C:\>arp -a
No ARP Entries Found
```

PCD

```
C:\>arp -a
No ARP Entries Found
```

PCE

```
C:\>arp -a
No ARP Entries Found
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- a. Click **PCC** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a

    Internet Address      Physical Address         Type
192.168.11.1             0002.4a00.0e91          dynamic
```

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time=1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a

    Internet Address      Physical Address         Type
192.168.11.1             0002.4a00.0e91          dynamic
192.168.11.112           0001.6462.0278          dynamic
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```

C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Reply from 192.168.4.110: bytes=32 time=19ms TTL=126
Reply from 192.168.4.110: bytes=32 time=17ms TTL=126
Reply from 192.168.4.110: bytes=32 time=15ms TTL=126
Reply from 192.168.4.110: bytes=32 time=13ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 19ms, Average = 16ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic

```

- e. Discuss the results you got from all the commands on PCC.
- The ARP entries are revealed in the ARP table only after PCC pings PCB, PCD, and PCE. No ARP entries are found before these ping activities. The ARP table displays entries once PCB, PCD, and PCE receive the ARP request from PCC and subsequently receive pings from PCB, PCD, and PCE.
- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```

RTA>enable
RTA#show arp

```

```

RTA>enable
RTA#show arp

```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.11.1	-	0002.4A00.0E91	ARPA	FastEthernet1/0
Internet	192.168.11.11	3	00D0.D39A.C0D9	ARPA	FastEthernet1/0
Internet	192.168.33.1	-	000C.CF0C.593A	ARPA	FastEthernet0/0
Internet	192.168.33.22	3	0060.47EA.A746	ARPA	FastEthernet0/0

- g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```

RTB>enable
RTB#show arp

```

```

RTB>enable
RTB#show arp

```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.4.1	-	0001.977A.B614	ARPA	FastEthernet0/0
Internet	192.168.4.110	1	0060.702D.7C08	ARPA	FastEthernet0/0

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.6462.0278   DYNAMIC Fa1/1
1       0002.4a00.0e91   DYNAMIC Fa0/1
1       000c.8546.7d85   DYNAMIC Fa1/1
1       00d0.d39a.c0d9   DYNAMIC Fa2/1
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB# show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a   DYNAMIC Fa0/1
1       0060.47ea.a746   DYNAMIC Fa1/1
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
```

```
SWC#show arp  
  
SWC#show mac-address-table
```

```
SWC>enable  
SWC#show arp  
  
SWC#show mac-address-table  
      Mac Address Table  
-----  
  
Vlan    Mac Address      Type      Ports  
----    -  
      1    0001.977a.b614    DYNAMIC   Fa0/1  
      1    0060.702d.7c08    DYNAMIC   Fa1/1
```

Switch

```
Switch>enable  
Switch#show arp  
  
Switch#show mac-address-table  
      Mac Address Table  
-----  
  
Vlan    Mac Address      Type      Ports  
----    -  
      1    0005.5e95.187b    DYNAMIC   Fa0/1
```

- d. Do switches use arp table? (Y/N)

No

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation.*

It is not part of NetComm syllabi, it is just for knowledge...

Switches employ a MAC address table (or CAM table) to map MAC addresses to specific switch ports. Without using IP addresses or ARP, switches can use this table to forward packets to the appropriate location at Layer 2. This is because they are level 2 switches and does not use the ARP table.

- f. What information does the command **show mac-address-table** gives?

The show mac-address-table command displays the MAC address table on a switch, which includes the following information: the MAC addresses that have been learned or manually configured, the VLAN associated with each MAC address, the type of entry (either DYNAMIC for learned addresses or STATIC for manually configured ones), and the physical switch port where the device with the corresponding MAC address is connected.

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. *Example: Lab4AliAhmad.pkt
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot shows the Cisco Packet Tracer interface. On the left, a network topology is displayed with a switch (S1) connected to a wireless router (WRS2). S1 has three subinterfaces: G0/0.10 (172.17.10.1/24), G0/0.20 (172.17.20.1/24), and G0/0.88 (172.17.88.1/24). PC1 is connected to S1 F0/11 (172.17.10.21/24, VLAN 10). PC2 is connected to S1 F0/18 (172.17.20.22/24, VLAN 20). WRS2 is connected to S1 F0/7 and has a wireless interface (172.17.88.25/24, VLAN 88). PC3 is connected to WRS2 (172.17.40.100/24). On the right, the 'PT Activity' window is open, titled 'Packet Tracer – Configuring Wireless LAN Access'. It shows an 'Addressing Table' with the following data:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

At the bottom of the PT Activity window, the 'Completion' status is shown as 100/100.

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

This screenshot shows the same network topology as the previous image. The 'PT Activity' window is open, showing a detailed scenario for configuring a wireless router. The scenario includes the following steps:

- Part 1: Configure a Wireless Router**
 - Step 1: Connect the Internet interface of WRS2 to S1.** Connect the WRS2 Internet interface to the S1 F0/7 interface.
 - Step 2: Configure the Internet connection type.**
 - Click WRS2 > GUI tab.
 - Set the Internet Connection type to Static IP.
 - Configure the IP addressing according to the Addressing Table.
 - Step 3: Configure the network setup.**
 - Scroll down to Network Setup. For the Router IP option, set the IP address to 172.17.40.1 and the subnet mask to 255.255.255.0.
 - Enable the DHCP server.

The PT Activity window also shows a progress bar for 'Time Elapsed: 00:00:00' and 'Completion: 100/100'. At the bottom of the Packet Tracer interface, a table shows the status of the network components:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
-	-	PC3	PC2	ICMP	0.000	N	0	(edit)	(delete)	(delete)
-	-	PC3	PC1	ICMP	0.000	N	1	(edit)	(delete)	(delete)